

Research References & Technical Foundation

A concise map from established computer-vision and geospatial research to AeroForge3D's implemented, demonstrated, and planned system stages.

Research, engineering and validation evidence

EXECUTIVE VIEW / 02

One pipeline, with different levels of proof

Research supports a coherent design. Repository code and run evidence determine what AeroForge3D has actually implemented.

AeroForge3D combines deterministic video preparation, optional timestamp-linked telemetry, a recoverable reconstruction task, map and model review, and persistent task measurements. The foundational ideas are established: local feature correspondences support camera pose and sparse geometry; multi-view stereo densifies that model; coordinate systems and independent checkpoints are essential for geospatial interpretation; and visualization, defect inference, and measurement are separate downstream functions. [T1-T8, T12-T14]

The evidence supports a working prototype, not an accuracy-certified inspection system. The default selector is deterministic 1 fps; learned feature support exists as an experiment and is not the production default. A single georeferenced reconstruction generated a textured model, point cloud, orthophoto, and DSM, but visible gaps and missing independent survey checkpoints limit claims. Gaussian Splatting, automated crack/anomaly detection, and low-light performance are planned or unvalidated.

Engineering principle: a method paper informs a design choice; a passing implementation test proves behavior under that test; an independently surveyed reference is needed to validate metric output. These evidence levels stay separate throughout this document.

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01 / SYSTEM PIPELINE

Research informs each stage; project status tells the present

The implementation path below uses AeroForge3D's own workflow terms. The literature references describe the method family, not a claim that the project implements one named academic pipeline.

IMPLEMENTED	Video preflight and frame preparation FFprobe/FFmpeg inspection; deterministic one-frame-per-second default; ordered manifest and frame hashes.
IMPLEMENTED	Optional SRT association Valid DJI SRT records are matched to frame timestamps; EPSG:4326 positioning inputs are serialized and read back.
EXPERIMENTAL	Frame selection Default R1 remains fixed 1 fps. Classical and LightGlue-assisted alternatives are experimental; no quality or speed gain is validated.
SINGLE RUN	Feature matching and camera reconstruction One complete 194-image task reported all images registered; output coverage is fragmented.
DEMONSTRATED / LIMITED	Dense products and geospatial outputs Point cloud, mesh, orthophoto, and DSM generated. Telemetry alignment is not independent accuracy validation.
DEMONSTRATED	Interactive review and saved measurements Model, point cloud, map tiles, distance / area / DSM-volume records, and GeoJSON export verified.
PLANNED	Gaussian Splats and automated inspection Not among current outputs. Crack/anomaly detection and structural diagnosis are planned; low-light not validated.

Figure 1. Evidence status across the current AeroForge3D path, from capture to downstream interpretation.

Status terms: Implemented = present in the project; Demonstrated = exercised in a traceable run; Experimental = available in a separate research path but not promoted; Planned = not yet implemented or validated. Sources: README.md, docs/PRODUCTION_BASELINE.md, selected source modules, and the 29 September 2026 run.

02 / TECHNICAL FOUNDATIONS

From image evidence to spatial products

The methods form a dependency chain. Errors and missing evidence at an upstream stage constrain what can be claimed downstream.

Stage	Technical role	Research anchor	AeroForge3D relationship
Feature detection / matching	Find the same scene structure across overlapping views; reject geometrically inconsistent pairs.	Scale-invariant local features [T3]; learned local matching [T15].	LightGlue/SIFT adapter exists for experimental R3 scoring; default production selection does not use it. Matching accuracy has not been isolated as a project benchmark.
Structure-from-Motion	Estimate camera parameters and poses while triangulating a sparse scene and refining reprojection error.	Incremental registration, triangulation, bundle adjustment [T1]; practical SfM/MVS stages [T2].	A reconstruction task demonstrated camera registration and sparse points. No claim that AeroForge3D implements COLMAP.
Multi-view stereo	Use calibrated views to estimate dense depth and fuse scene surfaces.	Pixelwise view selection, depth/normal estimation and fusion [T4].	Dense point cloud, mesh, DSM, and orthophoto were produced in the demonstrated run; coverage is visibly incomplete.
Georeferencing	Transform a local reconstruction into a named horizontal CRS and, separately, a meaningful vertical reference.	RTK direct-georeferencing assessment [T6]; coordinate operation concepts [T8].	SRT-to-frame alignment and EPSG:4326 input bundle implemented. Output CRS is recorded, but no surveyed checkpoints or vertical datum were supplied.
Output accuracy assessment	Compare positions to independent checkpoints with defined CRS, units, uncertainty, and statistics.	ASPRS positional accuracy standard [T7].	Planned validation method. Current software-reported residuals are not independently verified accuracy.

Interpretation guardrail. A low pixel reprojection error measures image-space fit. It does not by itself establish geospatial position, scale, vertical datum, completeness, or crack-width accuracy. [T1, T6, T7]

The UAV photogrammetry review identifies image quality, scene characteristics, and processing tools as determinants of product quality; the project run displays that dependence concretely through fragmented output on low-texture ground and vegetation. [T5]

03 / RESEARCH-TO-SYSTEM MAP

What each source contributes - and how AeroForge3D uses it

“Implements” below refers to a project behavior supported by local files or run artifacts. “Inspired by” means a method informs design but is not claimed as an exact implementation.

Research / contribution	System element informed	Relationship and current proof
SfM: image registration, triangulation, bundle adjustment [T1]	Matching -> camera poses -> sparse structure	Implements the workflow through an integrated reconstruction task; one dataset reports 194/194 registered images. Coverage and absolute accuracy remain limited.
COLMAP technical workflow [T2]	Reference architecture for sparse then dense reconstruction	Reference only. It explains stage boundaries; the project does not claim to use COLMAP.
SIFT and learned matching [T3,T15]	Feature support and experimental frame ranking	SIFT + LightGlue code and a pinned model are present for an experimental path. The R3 trial did not complete a comparable full reconstruction.
MVS view selection / depth fusion [T4]	Dense point cloud, mesh, DSM, orthophoto	Method-family foundation; outputs demonstrated in one run. No controlled dense-geometry comparison.
UAV SfM/MVS quality review [T5]	Capture assumptions and limitation disclosures	Informs overlap, image quality, scene texture, and tool sensitivity; not a numerical result transferable to AeroForge3D.
RTK direct georeferencing and accuracy assessment [T6]	Telemetry inputs, CRS handling, checkpoint plan	SRT associations implemented. Research supports checking output against surveyed checkpoints; none were used in the current run.
ASPRS positional accuracy standard [T7] + PROJ operations [T8]	Accuracy report design; CRS and transformation vocabulary	Informs the future validation protocol. No standards compliance or PROJ implementation is claimed.
3D Gaussian Splatting [T9]	Future visualization branch	Inspired by a rendering representation initialized from calibrated views; planned only. It is not a current product output.
Low-light studies / enhancement [T10,T11]	Future capture and preprocessing experiments	Literature indicates lighting and blur affect image quality. No AeroForge3D night test or enhancement path has been validated.
Infrastructure and crack inspection reviews [T12,T13]	Future defect triage, source-image linkage, and review	Inspection/detection/diagnosis remain planned. A 3D model is not an inspection conclusion.
Photogrammetric crack gauge [T14]	Future 3D defect measurement	Research precedent only; crack detection and width measurement are not implemented in AeroForge3D.

The R3 selector experiment is kept separate from the stable deterministic R1 baseline. Its incomplete run cannot support a claim of better efficiency, registration, or geometry.

04 / DOWNSTREAM CAPABILITIES

Rendering is not measurement; inspection is not diagnosis

A technical foundation is strongest when it defines the boundary between a representation and the decision made from it.

Gaussian Splatting

The original 3D Gaussian Splatting work uses an explicit Gaussian scene representation initialized from sparse points and optimized for high-quality, real-time novel-view synthesis. That makes it a plausible future visualization branch after camera calibration. It does not, on its own, guarantee a survey surface, an orthometric height, or measurement precision. Commercial products document splat output alongside conventional meshes and mapping products, so splats are a representation choice rather than a defensible market distinction. [T9, M5-M9]

Low-light and difficult imagery

A UAV day/night study reports sensitivity of camera calibration to poor lighting and identifies exposure, image sharpness, lighting uniformity, and control-point distribution as quality factors. Enhancement research such as Zero-DCE can adjust image luminance, but its vision-quality objective is not photogrammetric registration or geometry. AeroForge3D has no validated low-light result; any preprocessing must be tested for effects on feature matching and surface accuracy, not judged only by image appearance. [T10, T11]

Crack and structural inspection

Crack detection requires defect-specific image acquisition, labels, evaluation, and often scale calibration. Infrastructure reviews distinguish visible-damage localization from structural diagnosis and monitoring; recent UAV photogrammetry work evaluates crack width against physical measurement instruments. These sources motivate an eventual inspection layer, but they do not validate AeroForge3D detections. [T12-T14]

Measurement

Distance and area depend on correctly scaled geometry and coordinate handling. Volume additionally depends on surface semantics, NoData treatment, and a defensible vertical unit and datum. Current task-associated records demonstrate persistence and export, while the selected paths and the DSM itself have not been checked against surveyed or physical truth. ASPRS guidance provides a relevant framework for a future accuracy report, not a certification already achieved. [T7, T8]

System boundary. A mesh, point cloud, ortho, DSM, and saved geometry are intermediate evidence products. “Crack found,” “structure safe,” “survey accurate,” and “change measured” each require a separate validation design.

05 / IMPLEMENTATION BOUNDARIES

Technologies in the present system, and the open proof gates

This inventory names product-owned workflow components and formats visible in the repository and run evidence.

Layer	Technology or artifact	Current role / status
Preparation	Python 3.12 CLI and companion API; FFmpeg / FFprobe	Video inspection, deterministic frame extraction, manifests, SRT parsing, frame association, and geolocation input generation are implemented. Environment inventory records FFmpeg 9.0.1; run-specific full decode returned exit 0.
Orchestration	Containerized Linux workspace; task records and recovery logs	Preflight, task submission, status, retry / resume, output inventory, and failure records were demonstrated on the 29 September run.
Review	Leaflet map workspace and Potree 3D viewer	Orthophoto tiles, textured GLB, and LAZ point cloud rendered; map measurement records survived reload.
Spatial products	GLB, LAZ, GeoTIFF, GeoJSON	Textured mesh, point cloud, orthophoto, DSM, and measurement export are available for the demonstrated dataset.
Selection research	Classical blur / redundancy / feature proxies; SIFT + LightGlue experiment	Experimental branch only; deterministic R1 remains default. No promotion based on controlled quality or runtime evidence.
Planned extensions	Gaussian Splat output; crack/anomaly detection; structural diagnosis	Not validated outputs. Low-light is untested; accuracy testing needs independent control and check data.

Evidence gates before stronger claims

Claim to support	Minimum next evidence
Survey-grade / sub-meter positioning	CRS and vertical datum declared; independent checkpoints withheld from adjustment; suitable accuracy statistic, sample count, residuals, and uncertainty.
Better frame selection	Paired locked-input runs on multiple sites; same reconstruction options and resources; output completeness and checkpoint geometry alongside runtime.
Faster than real time / under 10 minutes	Full 10-minute source recordings; wall-clock boundary includes preparation, queue, processing, exports, retries, and usable outputs; cold repeats on named hardware.
Low-light robustness	Paired lighting and shutter conditions with blur/exposure measures; report image registration, coverage, and independent geometry.
Automated defect detection / structural assessment	Annotated task dataset, expert-reviewed labels, sensitivity/precision, localization or width error, and engineering review boundaries.

Current limitations and exact one-run metrics are reported separately in 03_Real_World_Results_Validation.pdf.

SELECTED REFERENCES / TECHNICAL FOUNDATION

Primary papers, standards, and documentation

Curated references selected for direct relevance to image reconstruction, geospatial validation, rendering, inspection, and measurement.

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[T7] ASPRS. (2024). *Positional Accuracy Standards for Digital Geospatial Data*, Edition 2, Version 2.0.
ASPRS / current standards page

[T8] PROJ contributors. *Coordinate operations and geodetic transformations*, PROJ documentation.
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[T12] Ai, D., Jiang, G., Lam, S.-K., He, P., & Li, C. (2023). Computer vision framework for crack detection of civil infrastructure - A review. *Engineering Applications of Artificial Intelligence*, 117, 105478.
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[T13] Spencer, B. F. Jr., Hoskere, V., & Narazaki, Y. (2019). Advances in Computer Vision-Based Civil Infrastructure Inspection and Monitoring. *Engineering*, 5(2), 199-222.
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[T14] Gil-Docampo, M. L., et al. (2025). Photogrammetric-based gauge for remote monitoring and measurement of inaccessible cracks using unmanned aerial vehicles. *Structural Health Monitoring*.
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[T15] Lindenberger, P., Sarlin, P.-E., & Pollefeys, M. (2023). LightGlue: Local Feature Matching at Light Speed. *ICCV*, 17627-17638.
doi:10.1109/ICCV51070.2023.01616

Use note: citations support technical concepts and validation methods. They do not imply that AeroForge3D uses a named library or achieves a paper's published result. Project implementation status is sourced from local repository evidence.